

Problem Statement: Vector Multiplication

```
%%writefile matrix_mul_input.cu

#include <stdio.h>

#define N 3 // Change size if needed

__global__ void matrixMultiply(int *A, int *B, int *C) {
    int row = threadIdx.y;
    int col = threadIdx.x;

    int sum = 0;

    for (int k = 0; k < N; k++) {
        sum += A[row * N + k] * B[k * N + col];
    }

    C[row * N + col] = sum;
}

int main() {
    int A[N][N], B[N][N], C[N][N];

    int *d_A, *d_B, *d_C;

    // Take input for Matrix A
    printf("Enter elements of Matrix A:\n");
    for(int i=0;i<N;i++){
```

```
    for(int j=0;j<N;j++){  
        scanf("%d", &A[i][j]);  
    }  
}
```

```
// Take input for Matrix B  
printf("Enter elements of Matrix B:\n");  
for(int i=0;i<N;i++){  
    for(int j=0;j<N;j++){  
        scanf("%d", &B[i][j]);  
    }  
}
```

```
// Allocate memory on GPU  
cudaMalloc((void**)&d_A, N*N*sizeof(int));  
cudaMalloc((void**)&d_B, N*N*sizeof(int));  
cudaMalloc((void**)&d_C, N*N*sizeof(int));
```

```
// Copy data to GPU  
cudaMemcpy(d_A, A, N*N*sizeof(int), cudaMemcpyHostToDevice);  
cudaMemcpy(d_B, B, N*N*sizeof(int), cudaMemcpyHostToDevice);
```

```
dim3 threads(N, N);
```

```
// Kernel launch  
matrixMultiply<<<1, threads>>>(d_A, d_B, d_C);
```

```
// Copy result back
```

```

cudaMemcpy(C, d_C, N*N*sizeof(int), cudaMemcpyDeviceToHost);

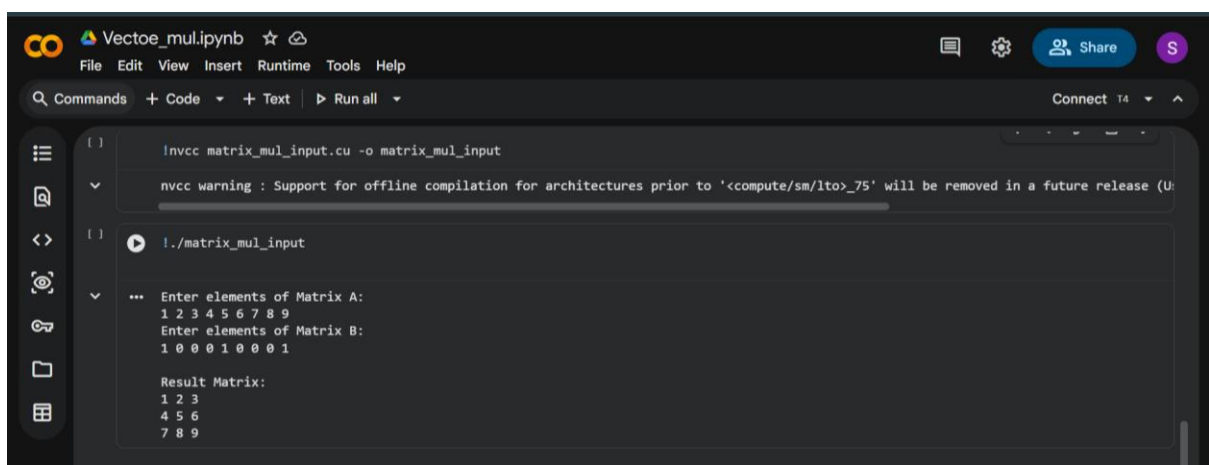
// Print result
printf("\nResult Matrix:\n");
for(int i=0;i<N;i++){
    for(int j=0;j<N;j++){
        printf("%d ", C[i][j]);
    }
    printf("\n");
}

// Free memory
cudaFree(d_A);
cudaFree(d_B);
cudaFree(d_C);

return 0;
}

```

OUTPUT:



```

Vectoe_mul.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Connect T4

[]
!nvcc matrix_mul_input.cu -o matrix_mul_input
nvcc warning : Support for offline compilation for architectures prior to '<compute/sm/lto>_75' will be removed in a future release (U:

[]
!./matrix_mul_input
... Enter elements of Matrix A:
1 2 3 4 5 6 7 8 9
Enter elements of Matrix B:
1 0 0 0 1 0 0 0 1
Result Matrix:
1 2 3
4 5 6
7 8 9

```